

Noetic Math Contest: National Honor Roll Practice (Grade 3)

Name: _____ Time Limit: 45 Minutes

1. An eraser covers the same digit in this correctly solved addition problem. What digit is hidden?

$$\begin{array}{r} (?) 7 \\ + 6 (?) \\ \hline 111 \\ \hline \end{array}$$

Answer: _____

2. Victor sends 8 texts a day to each of his friends. If he sends a total of 144 texts in 3 days, how many friends does he have?

Answer: _____ friends

3. How many 3-digit numbers can you make using only the digits 0, 5, and 8? You can use each digit as many times as you want (e.g., 550, 888, 580).

Answer: _____ numbers

4. Jeff left home on a Tuesday. He returned exactly 50 days later. What day of the week did he return?

Answer: _____

5. A balance scale is level. On the left side are three 120g weights. On the right side are two identical books and one 40g ball. How much does each book weigh?

Answer: _____ g

6. A 24-meter rope is cut into two pieces. One piece is three times as long as the other. How long is the longer piece?

Answer: _____ m

7. Find the sum of the 9th and 10th numbers in this pattern: \$4, 11, 18, 25, ...\$

Answer: _____

8. In a line of students, Sarah is 12th from the front. Tim is 15th from the back. There are 4 students standing between Sarah and Tim. If Sarah is closer to the front than Tim, how many students are in the line?

Answer: _____ students

9. A large cube is made of $3 \times 3 \times 3$ small cubes. If the outside of the large cube is painted red, how many small cubes have exactly two sides painted red?

Answer: _____ cubes

10. If 2 cats can catch 2 mice in 2 minutes, how many cats are needed to catch 10 mice in 10 minutes?

Answer: _____ cats

11. One apple costs the same as 3 plums. Two apples and 2 plums together cost \$16. How much does one apple cost?

Answer: \$_____

12. A clock shows 3:00. If the minute hand moves 120 degrees clockwise, how many minutes have passed?

Answer: _____ minutes

13. I am a 2-digit number. If you reverse my digits, the new number is 18 more than my original number. The sum of my digits is 10. What number am I?

Answer: _____

14. In a class of 30 students, 20 like Pizza and 15 like Tacos. Every student likes at least one. How many students like both?

Answer: _____ students

15. A rectangle is made by putting two identical squares side-by-side. The perimeter of the rectangle is 30 cm. What is the area of the rectangle?

Answer: _____ sq cm

16. How many times does the digit '5' appear in the numbers from 10 to 99?

Answer: _____ times

17. A snail is at the bottom of a 20ft well. Each day it climbs up 5ft, but each night it slides down 3ft. On which day will it reach the top?

Answer: Day _____

18. Find different digits A and B that make this true: $AA + BB = 121$.

Answer: A = _____, B = _____

19. A bag contains 10 red socks and 10 blue socks. What is the smallest number of socks you must pull out (without looking) to be sure you have two different colors?

Answer: _____ socks

20. Four friends ran a race. A finished before B. C finished after D. B finished before D. Who came in last place?

Answer: _____